Multivariate Calculus Lecture Notes
Section 14.7 – Optimization in Several Variables

I. Definition of local extrema
   a. The function f has a *local minimum* or *relative minimum* at (a, b) if $f(x, y) \geq f(a, b)$ for all (x, y) in an open disk containing (a, b).
   b. The function f has a *local maximum* or *relative maximum* at (a, b) if $f(x, y) \leq f(a, b)$ for all (x, y) in an open disk containing (a, b).

II. Finding local extrema
   a. Critical points
      i. Let f be defined on an open region containing (a, b). The point (a, b) is a *critical point* of f if one of the following is true:
         1. $f_x(a, b) = 0$ and $f_y(a, b) = 0$ (i.e., $\nabla f(x, y) = \mathbf{0}$)
         2. $f_x(a, b)$ or $f_y(a, b)$ does not exist (i.e., $\nabla f(x, y)$ is undefined)
      ii. Fermat's Theorem: Local extrema occur only at critical points.
      iii. Why could we get a max or min if the gradient doesn't exist?
         - If $f_x(a, b)$ or $f_y(a, b)$ does not exist then we have some different situations, one of which is a cusp:

      iv. If only one of $f_x(a, b)$, $f_y(a, b)$ is zero then there may not be a local extremum.
         1. Example: $z = y$.
         2. $f_x(x, y) = 0$ for all x and y, but there are no local extrema anywhere on this plane.
      v. This does not mean that every critical point is a local extremum.
         1. How could we have $f_x(a, b) = 0$ and $f_y(a, b) = 0$ but (a, b) not be a local extremum?
         2. The classic example of this is a hyperbolic paraboloid (i.e., a saddle).
   b. Examples: Find the critical points of $f(x, y) = x^2 - 3xy - y^2 + 6x$.
      i. $f_x(x, y) = 2x - 3y + 6$, $f_y(x, y) = -3x - 2y$
      ii. Thus we must simultaneously solve $2x - 3y + 6 = 0$ and $-3x - 2y = 0$.
      iii. Multiply the first equation by 3 and the second by 2: $6x - 9y + 18 = 0$, $-6x - 4y = 0$.
      iv. Add: $-13y + 18 = 0$. Thus $y = 18/13$.
      v. And therefore $-3x - 2(18/13) = 0 \implies x = -12/13$.
      vi. So the only critical point is $(-12/13, 18/13)$.
      vii. Is this a local maximum or a local minimum? We don't know yet.

III. The second partials test
   a. With single-variable calculus we can determine whether a critical point is a relative maximum, a relative minimum, or neither by the use of the second derivative (provided the first derivative exists).
   b. Similarly, in multivariable calculus we have the *second partials test* (again, provided both first partials exist).
      i. Given a point (a, b) for which $f_x(a, b)$ and $f_y(a, b)$ are both zero, calculate the quantity $D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$.

ii. The value $D(a,b)$ is called the *discriminant* or *Hessian* of f at (a,b) and can be thought of as the value of the following determinant: $\begin{vmatrix} f_{xx}(a,b) & f_{xy}(a,b) \\ f_{yx}(a,b) & f_{yy}(a,b) \end{vmatrix}$.

iii. Interpreting the value of D
1. If $D(a,b) > 0$ and $f_{xx}(a,b) > 0$, then f has a local minimum at (a,b).
2. If $D(a,b) > 0$ and $f_{xx}(a,b) < 0$, then f has a local maximum at (a,b).
3. If $D(a,b) < 0$ then f has a saddle point at (a,b).
4. If $D(a,b) = 0$ then the test fails.

c. Explanation of the second partials test

i. The text has a derivation at the end of the section.

ii. If $f_{xx}(a,b)$ and $f_{yy}(a,b)$ have different signs, then the graph of f at (a,b) is concave up in one direction and concave down in the other.
1. So you would expect to have a saddle point at (a,b).
2. Also, $f_{xx}(a,b)$ and $f_{yy}(a,b)$ having different signs means that $D(a,b)$ would be negative, so we would have a saddle according to the second partials test.

iii. If $f_{xx}(a,b)$ and $f_{yy}(a,b)$ are both positive, then the graph of f at (a,b) is concave up in both x and y directions.
1. So it would seem that there would be a local minimum at (a,b).
2. However, you can have a graph like this: $f(x,y) = x^2 - 5xy + y^2$.
   - Show on Mathematica.
   - $f_x(x,y) = 2x - 5y$, so $f_x(0,0) = 0$.
   - $f_y(x,y) = -5x + 2y$, so $f_y(0,0) = 0$.
   - Also, $f_{xx}(x,y) = 2$, $f_{yy}(x,y) = 2$.
   - No local min at $(0,0)$, though.
3. So it's not enough, for a local minimum, to have $f_{xx}(a,b)$ and $f_{yy}(a,b)$ both positive.
4. The graph is also twisting in both the x and y directions.

iv. The quantity $D(a,b) = f_{xx}(a,b)f_{yy}(a,b) - [f_{xy}(a,b)]^2$ basically says:
1. If the twisting effect from $f_{xy}(a,b)$ and $f_{yx}(a,b)$ is stronger than the concavity effect from $f_{xx}(a,b)$ and $f_{yy}(a,b)$ then we have a saddle.
2. If the concavity effect is stronger, then we have a local extremum.
   - If both $f_{xx}(a,b)$ and $f_{yy}(a,b)$ are positive, then we have a local minimum.
   - If both $f_{xx}(a,b)$ and $f_{yy}(a,b)$ are negative, then we have a local maximum.
3. If the concavity and twisting effects cancel each other out (or are zero to begin with), then the slope of the surface at (a,b) is flat in some direction, and we could have anything.

d. Examples

i. Finish the example of $f(x,y) = x^2 - 3xy - y^2 + 6x$: Is $(-12/13, 18/13)$ a local maximum or a local minimum – or neither?
1. We need the second partials.
   - $f_{xx}(x,y) = 2$, $f_{yy}(x,y) = -3$
   - $f_{yy}(x,y) = -2$
2. So $D(-12/13, 18/13) = -4 - 9 = -13$.
3. So $(-12/13, 18/13)$ is actually a saddle point.

ii. Find and identify the critical points of the function given by $f(x,y) = x^4 + y^4 - 4xy + 1$.
   1. We need the first partial derivatives.
      - $f_x(x,y) = 4x^3 - 4y$.
      - $f_y(x,y) = 4y^3 - 4x$.
   2. We need the critical points.
      - $0 = 4x^3 - 4y \implies y = x^3$.
      - $0 = 4y^3 - 4x \implies x = y^3$.
      - Thus we have $y = y^9 \implies y(y^8 - 1) = 0 \implies y = 0, y = 1$ or $y = -1$ (on the real line, anyway).
      - If $y = 0$, $x = 0$. If $y = 1$, then $x = 1$. If $y = -1$, then $x = -1$.
      - So our three critical points are $(0,0)$, $(1,1)$, and $(-1,-1)$.
   3. Now, the second partial derivatives.
      - $f_{xx}(x,y) = 12x^2$, $f_{xy}(x,y) = -4$.
      - $f_{yy}(x,y) = 12y^2$, $f_{yx}(x,y) = -4$.
   4. And now the second partials test.
      - We have $D(0,0) = 0 - 16 = -16$. Thus f has a saddle point at $(0,0)$.
      - We have $D(1,1) = 144 - 16 = 128$. Since $f_{xx}(1,1) = 12$, f has a local minimum at $(1,1)$.
      - We have $D(-1,-1) = 144 - 16 = 128$. Since $f_{yy}(1,1) = 12$, f has a local minimum at $(-1,-1)$.
   5. Plot function on Mathematica.

IV. Constrained optimization

a. Extreme Value Theorem

i. If f is a continuous function of x and y on a closed and bounded domain in the xy-plane, then f has at least one absolute minimum and one absolute maximum value on that region.

ii. Why does *bounded* matter?
   1. Remember that a region is *bounded* if you can draw a circle of finite radius around the domain.
   2. If the function is not bounded on the region then it might take on infinitely large or small values.
   3. Example: $z = 1 - x - y$ on the domain $\mathbb{R} \times \mathbb{R}$
      - This is a plane that extends to arbitrarily large and small values.
      - However, if the domain were bounded (and closed) then maxima and minima would exist.

iii. Why does *closed* matter?
   1. Remember that a region is *closed* if it contains its boundary points.
   2. If the function is not closed on the region then you can get arbitrarily close to the boundary, but you don't actually touch it.
   3. Example: $f(x,y) = 4 - x^2 - y^2$ on the domain $x^2 + y^2 < 4$.
      - Graphically:

- This function does not include the xy plane, and thus the minimum does not exist.

iv. Why does *continuous* matter?

1. If the function is not continuous, then it might "jump" in the domain.
2. Example: $f(x,y) = \begin{cases} 4 - x^2 - y^2, & x^2 + y^2 < 4; \\ 4, & x^2 + y^2 = 4. \end{cases}$
   - Graphically:

   - The domain is closed and bounded.
   - However, there is no minimum value because f is not continuous.

b. Finding absolute extrema of a continuous function f on a closed and bounded domain D

i. Absolute extrema occur at two places:

1. Critical points of f in D
2. Boundary points of D (illustrate this on Mathematica with elliptic paraboloid $z = x^2 + y^2$ on the domain $[-1,1] \times [-1,1]$).

ii. To find absolute extrema,

1. Find critical points of f in D.
2. Find the extreme values of f on the boundary of D.
3. The largest point of these is the absolute maximum value of f on D. The smallest of these is the absolute minimum value of f on D.
4. You don't need the second partials test!

iii. Example: Find the absolute maximum and minimum values of the function $f(x,y) = x^2 - 2xy + 2y$ on the domain $D = \{(x,y) | 0 \le x \le 3, 0 \le y \le 2\}$.

1. D is the following rectangle:

2. First, find critical values of f on D.
   - $f_x(x,y) = 2x - 2y$, $f_y(x,y) = -2x + 2$
   - Set the partials equal to 0 and solve to get $(1,1)$ as the only critical point of f.
   - Is $(1,1)$ in D? Yes.
3. Next, find the extreme values of f on the boundary of D.
   - On the boundary $y = 0$ we have the function $f(x) = x^2$.
     - As x runs from 0 to 3 $f(x) = x^2$ is an increasing function.
     - Thus the minimum value is at $(0,0)$ and the maximum value is at $(3,0)$.
   - On the boundary $x = 0$ we have the function $f(y) = 2y$.
     - As y runs from 0 to 2 $f(y) = 2y$ is an increasing function.
     - Thus the minimum value is at $(0,0)$ and the maximum value is at $(0,2)$.
   - On the boundary $y = 2$ we have the function $f(x) = x^2 - 4x + 4$.
     - This is a parabola.
     - Differentiate: $f'(x) = 2x - 4$.

  - Set to 0 and solve: $0 = 2x - 4 \implies x = 2$.
  - So we have the point $(2, 2)$.
  - We also need to check the boundary points $(0, 2)$ and $(3, 2)$.
- On the boundary $x = 3$ we have the function $f(y) = 9-6y+2y = 9-4y$.
  - As y runs from 0 to 2 $f(y) = 9 - 4y$ is a decreasing function.
  - Thus the minimum value is at $(3, 2)$ and the maximum value is at $(3, 0)$.

4. Finally, check all critical points and extreme value points on the boundary of D.
   - $f(1, 1) = 1$
   - $f(0, 0) = 0$
   - $f(3, 0) = 9$
   - $f(0, 2) = 4$
   - $f(3, 2) = 1$
   - $f(2, 2) = 0$
5. Thus the absolute maximum of f on D occurs at $(3, 0)$, with a value of 9. The absolute minimum of f on D occurs at both $(0, 0)$ and $(2, 2)$, with a value of 0.

Homework: Section 14.7 #3, 6, 7, 12, 19, 28, 35, 38, 47